a

ROC curves, IxIDN-Matched (n = 4,364; 50% positive)

True positive rate

False positive rate

AUROC

DisAtlas (full) 0.832

DisAtlas (benchmark diseases held out) 0.784

Claude (chat) 0.692

Claude (science agent) 0.685

Phylo Biomni 0.627

TxGNN 0.566

Gemini Pro 0.497

b

Ranking metrics (dot = value, bar = 95% CI, diseases resampled)

AUPRC

F1 at k = #pos.

Precision, top 10%

NPV, bottom 10%

DisAtlas (full) 0.842 0.760 0.956 0.867

DisAtlas (benchmark diseases held out) 0.789 0.719 0.913 0.851

Claude (chat) 0.739 0.628 0.933 0.635

Claude (science agent) 0.708 0.640 0.860 0.638

Phylo Biomni 0.664 0.596 0.830 0.500

TxGNN 0.606 0.541 0.773 0.530

Gemini Pro 0.536 0.497 0.631 0.489

random

c

Run-to-run agreement of pair scores

one point = one pair of runs; no box under 5 pairs; TxGNN (single run) not shown

Spearman ρ between two runs

DisAtlas (full)

DisAtlas (benchmark diseases held out)

Claude (chat)

Claude (science agent)

Phylo Biomni

Gemini Pro

d

Overlap of top-10% pairs (Jaccard; 436 of 4,364 per method)

	DisAtlas (full)	DisAtlas (benchmark diseases held out)	Claude (chat)	Claude (science agent)	Phylo Biomni	TxGNN	Gemini Pro
DisAtlas (full)	1.00	0.47	0.27	0.20	0.17	0.12	0.12
DisAtlas (benchmark diseases held out)	0.47	1.00	0.23	0.16	0.16	0.12	0.10
Claude (chat)	0.27	0.23	1.00	0.51	0.26	0.12	0.14
Claude (science agent)	0.20	0.16	0.51	1.00	0.26	0.09	0.15
Phylo Biomni	0.17	0.16	0.26	0.26	1.00	0.06	0.19
TxGNN	0.12	0.12	0.12	0.09	0.06	1.00	0.02
Gemini Pro	0.12	0.10	0.14	0.15	0.19	0.02	1.00

Jaccard index